

BELOZERSK, Russian Federation

WMO: 229390

Lat: 60.0203N

Lon: 37.7592E

Elev: 428

StdP: 14.47

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-13.0	-24.6	1.4	-19.4	-17.9	2.1	-12.8	23.3	23.6	21.6	23.3	3.2	170	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	82.2	67.5	78.5	65.7	75.1	63.8	69.9	79.3	67.5	75.6	65.4	72.9	7.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	98.9	74.5	64.3	91.5	72.4	62.2	85.0	69.7	34.1	79.1	32.1	75.6	30.5	73.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
21.7	19.1	16.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-25.2	86.8	7.4	3.4	-30.5	89.3	-34.9	91.3	-39.0	93.2	-44.4	95.7		
			-25.2	71.9	7.6	5.0	-30.6	75.5	-35.0	78.4	-39.3	81.3	-44.8	84.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	13.3	16.4	24.4	38.4	50.8	59.3	63.4	59.7	50.6	38.6	27.4	19.5
	DBStd	19.80	13.04	13.68	9.42	8.20	9.32	7.32	6.50	6.35	6.77	6.97	10.17	12.82
	HDD50	5411	1139	942	795	361	107	8	1	4	73	357	679	945
	HDD65	9757	1604	1362	1260	798	447	201	107	187	434	818	1129	1410
	CDD50	1248	0	0	0	13	132	287	418	305	89	4	0	0
	CDD65	121	0	0	0	0	7	30	59	24	1	0	0	0
	CDH74	990	0	0	0	1	80	273	441	193	2	0	0	0
	CDH80	220	0	0	0	0	12	53	107	48	0	0	0	0
Wind	WSAvg	7.6	7.8	8.5	8.8	7.7	7.6	6.9	5.9	6.5	7.0	7.8	8.3	8.6
Precipitation	PrecAvg	21.2	1.0	0.7	0.9	1.3	2.0	2.3	3.0	2.9	2.1	2.0	1.5	1.3
	PrecMax	28.9	2.1	1.8	2.3	3.2	3.4	6.5	6.5	7.7	4.4	4.4	3.3	2.5
	PrecMin	14.8	0.4	0.2	0.1	0.1	0.7	0.7	0.9	0.5	0.9	0.0	0.3	0.6
	PrecStd	3.5	0.4	0.4	0.6	0.7	0.8	1.2	1.4	1.7	0.9	0.9	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	37.7	46.8	70.0	81.4	85.5	87.9	86.1	73.4	60.0	47.5	40.9
		MCWB	34.0	35.0	38.5	55.0	61.8	69.8	71.5	71.3	62.2	53.8	45.9	39.4
	2%	DB	34.1	35.3	41.5	62.3	75.2	80.8	82.6	78.5	68.1	53.8	43.3	37.4
		MCWB	32.9	33.4	36.0	50.2	58.7	66.6	68.4	66.0	59.9	49.8	41.7	36.0
	5%	DB	33.0	33.8	38.3	56.5	70.1	77.0	78.8	74.6	63.9	51.1	40.9	35.3
		MCWB	32.0	32.2	34.0	46.2	56.7	64.4	67.6	64.4	57.7	48.2	39.9	34.1
	10%	DB	31.0	32.6	36.0	51.2	65.5	72.7	75.2	71.1	60.5	48.6	38.7	33.7
		MCWB	30.1	31.3	32.9	42.9	54.6	62.3	64.9	62.4	55.7	46.2	37.8	32.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	35.6	39.6	56.2	64.6	72.7	73.8	72.0	64.1	55.1	46.5	39.5
		MCDB	35.1	37.3	44.4	67.8	77.3	82.0	82.2	81.7	71.0	58.7	47.1	40.7
	2%	WB	33.2	33.6	36.8	51.1	61.0	68.5	70.4	67.9	61.0	51.2	42.0	36.3
		MCDB	33.9	34.7	40.5	60.6	71.0	77.3	80.3	75.9	66.2	53.2	43.0	37.1
	5%	WB	32.1	32.4	34.9	47.3	58.6	66.0	68.2	65.1	58.6	48.8	40.0	34.4
		MCDB	32.9	33.5	37.4	54.2	66.5	73.9	76.6	72.8	63.0	50.7	40.8	35.0
	10%	WB	30.0	31.3	33.4	43.8	56.2	63.6	66.2	63.2	56.1	46.3	37.9	32.8
		MCDB	31.1	32.6	35.8	50.4	63.6	71.0	73.5	69.9	59.9	48.3	38.7	33.5
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.8	13.7	15.7	17.1	16.2	15.9	15.3	12.2	8.4	6.8	8.4
		MCDBR	6.7	7.9	14.1	23.7	24.4	21.9	21.0	21.2	17.8	11.5	7.5	6.0
		MCWBR	6.9	7.4	10.0	14.1	11.2	10.7	10.1	10.2	10.2	8.6	7.1	5.7
	5% WB	MCDBR	6.7	7.7	10.6	21.6	19.5	19.5	19.1	19.2	15.4	10.5	7.4	5.9
		MCWBR	7.0	7.4	8.1	13.8	10.0	10.9	10.0	10.0	9.3	8.5	7.2	5.8
Clear-Sky Solar Irradiance	taub	0.262	0.306	0.328	0.370	0.358	0.356	0.389	0.382	0.346	0.327	0.286	0.240	
	taud	2.054	2.069	2.126	2.248	2.430	2.460	2.382	2.383	2.464	2.390	2.155	2.011	
	Ebn at Noon	154	210	249	262	276	278	264	256	248	211	145	113	
	Edh at Noon	15	26	33	36	32	32	34	31	24	19	13	10	
All-Sky Solar Radiation	RadAvg	102	312	760	1214	1607	1696	1647	1263	732	307	94	53	
	RadStd	14	71	103	144	104	103	204	86	72	50	17	8	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)		+1.24	N/A	N/A	N/A	N/A	N/A	-360	-439	N/A	N/A

Nomenclature: See separate page